BOTSWANA INTERNATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY

BOTSWANA INTERNATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY
(SEMESTER I, 2023/2024 FINAL EXAMINATIONS)

FACULTY/DEPARTMENT:	FET, Department of Electrical, Computer and Telecommunications Engineering
MODULE CODE:	EEEN 414
MODULE TITLE:	Digital Signal Processing
DATE:	28th November 2023
TIME:	14:00-16:00 (2 hours) plus 10 minutes reading time
VENUE(s):	122/001-002
EXAMINER(s):	Dr Bokamoso Basutli
ADDITIONAL MATERIALS:	**N/A**

Instructions to Candidates:

1. This question paper contains a total of **four (5)** pages (including this page) and the Appendix.
2. There are Two (2) sections in this question paper:
   Section I: Compulsory (75%)
   Section II: Choose only ONE question (25%)
   Total Marks: 100
3. Answer all questions in Section 1 and only ONE question in Section 2.
4. All work **including rough work** should be done in the answer booklet(s) provided. Cross out any rough work and indicate it as such to avoid it being marked.
5. Use only a Blue or Black pen for written answers and a pencil for all sketches/ drawings.
6. Ensure that your question paper contains all pages.
7. Only non-programmable calculators are allowed in this examination.
8. Possession of any other material not prescribed in these instructions including mobile phones/smart watches is a punishable offence as per the BIUST Misconduct Regulation.

function of the control unit.

SECTION 1: Answer ALL QUESTIONS.

1. (a) Compute the z-transform of (4 marks)

$$9(2)^n u[n] + 8(3)^n u[n] - 7(4)^n u[-n-1] + 6(5)^n u[-n-1]$$

(b) Determine the ROC of the z-transform of (4 marks)

$$x[n] = (-2)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$$

(c) Consider a z-transform $X(z)$ for which the partial fraction expansion of $\frac{X(z)}{z}$ is

$$\frac{X(z)}{z} = A_0 \frac{1}{z-0} + \sum_{i=1}^{N} A_i \frac{1}{z-p_i}$$

Multiplying both side by z gives:

$$X(z) = A_0 + \sum_{i=1}^{N} A_i \frac{z}{z-p_i}$$

where $\{p_i\}$ are the poles of $X(z)$ and $\{A_i\}$ are their associated residues. The (causal) inverse **unilateral** z-transform of $X(z)$ is given by

$$x[n] = A_0\delta[n] + \sum_{i=1}^{N} A_i p_i^n u[n].$$

In contrast, the inverse **bilateral** z-transform is given by

$$x[n] = A_0\delta[n] + \sum_{i=1}^{N} A_i \begin{cases} p_i^n u[n], & \text{if } \{|z| > |p_i|\}, \\ -p_i^n u[-n-1], & \text{if } \{|z| < |p_i|\} \end{cases}$$

because there are two choices (causal or anticausal) for each term in the partial fraction expansion, depending on the choice of region of convergence (ROC) for that term. This suggests that there are $2N$ possible inverse z-transforms, since the choice for each term can be made independently. But there are actually only $N+1$ possible inverse z-transforms, since the other choices lead to ROCs that are empty sets.

Use the explanation given above to compute all of the possible z-transform of under bilateral z-transform (12 marks)

$$X(z) = \frac{2z^2 - 2.5z}{z^2 - 2.5z + 1}$$

(d) Let $x_1[n] = \{3, 2, 4\}$ and $x_2[n] = \{3, 2, 4\}$. Compute the z-transforms of signals $y_1[n] = x_1[n-1]$ and $y_2[n] = x_2[n-1]$. (5 marks)

[25]

2. (a) Determine the zeros and poles and sketch the pole-zero plot on the z-plane (5 marks)

$$X(z) = z\frac{2z^2 - 2z + 1}{16z^3 + 6z + 5}$$

(b) The DTFT and its inverse are defined as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}$$

Compute the DTFT of (5 marks)

$$x[n] = \{3, 1, \underset{\uparrow}{4}, 2, 5\}$$

(c) Determine the Big $\mathcal{O}$ complexity of convolution sum. Show your workings. (4 marks)

(d) Determine the Big $\mathcal{O}$ complextity of DFT under straightforward implementation. Show your workings. (4 marks)

(e) The DFT is defined using the following analysis and synthesis equations.

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi k\frac{n}{N}}, \quad k = 0, 1, \ldots, N-1$$

$$x[n] = x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]e^{j2\pi k\frac{n}{N}}, \quad n = 0, 1, \ldots, N-1$$

(a) Prove that the IDFT will return $x[n]$ (3 marks)

(b) Consider the DFT of a length-6 sequence $x[n]$ as

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$$X[k] = \begin{cases} 4, & k = 0, \\ 2, & 1 \leq 5. \end{cases}$$

Compute $x[n]$ (4 marks)

[25]

3. Consider the pulse sequence, where $N \leq L$,

$$x[n] = u[n] - u[n-L] = \begin{cases} 1, & 0 \leq n \leq L-1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the DTFT of $x[n]$ and sketch its magnitude spectrum for $0 \leq \omega \leq 2\pi$ assuming $L = 10$ (10 marks)

(b) Determine the DFT of $x[n]$ by leveraging from Question 3(a) solution and sketch its magnitude spectrum for $0 \leq \omega \leq 4\pi$. (15 marks)

[25]

SECTION 2: Answer ONLY ONE QUESTION.

4. (a) Determine the DTFT of the exponential sequence $x[n] = a^n u[n]$. As a cautious first step, find out what conditions on a ensure that $x[n]$ is absolutely summable. Next, determine $X(e^{j\omega})$. (15 marks)

   (b) Determine the DTFT of the odd sequence $x[n] = -2\delta[n+2] + \delta[n+1] - \delta[n-1] + 2\delta[n-2]$. (10 marks)

[25]

5. Given the two signals

$$x_1[n] = \{\underset{\uparrow}{2}, 1, 4, 3\}$$

$$x_2[n] = \{\underset{\uparrow}{5}, 3, 2, 1\}$$

compute the cyclic convolution of the two signals by

   (a) applying the DFT method (15 marks)

   (b) applying the aliasing of the linear convolution method. (10 marks)

[25]

End of Questions

Appendix 1: z-Transform Pairs

z-Transform Pairs

	$x[n]$		$X(z) = \mathcal{Z}[x[n]]$	ROC
1	$\delta[n]$	$\leftrightarrow$	1	All z
1a	$\delta[n-m]$	$\leftrightarrow$	z^{-m}, if $m \geq 0$	$z \neq 0$
2	$u[n]$	$\leftrightarrow$	$\frac{z}{z-1}$	$\lvert z\rvert > 1$
3	$n\,u[n]$	$\leftrightarrow$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
3a	$n^2\,u[n]$	$\leftrightarrow$	$\frac{z(z+1)}{(z-1)^3}$	$\lvert z\rvert > 1$
4	$a^n\,u[n]$	$\leftrightarrow$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
4a	$a^{n-1}\,u[n-1]$	$\leftrightarrow$	$\frac{1}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
4b	$na^n\,u[n]$	$\leftrightarrow$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
4c	$(n-1)a^{n-2}\,u[n-2]$	$\leftrightarrow$	$\frac{1}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
4d	$n^2a^n\,u[n]$	$\leftrightarrow$	$\frac{az(z+a)}{(z-a)^3}$	$\lvert z\rvert > \lvert a\rvert$
5	$\sin(\Omega n)\,u[n]$	$\leftrightarrow$	$\frac{z\sin(\Omega)}{z^2-2z\cos(\Omega)+1}$	$\lvert z\rvert > 1$
5a	$a^n\sin(\Omega n)\,u[n]$	$\leftrightarrow$	$\frac{az\sin(\Omega)}{z^2-2az\cos(\Omega)+a^2}$	$\lvert z\rvert > \lvert a\rvert$
6	$\cos(\Omega n)\,u[n]$	$\leftrightarrow$	$\frac{z^2-z\cos(\Omega)}{z^2-2z\cos(\Omega)+1}$	$\lvert z\rvert > 1$
6a	$a^n\cos(\Omega n)\,u[n]$	$\leftrightarrow$	$\frac{z^2-az\cos(\Omega)}{z^2-2az\cos(\Omega)+a^2}$	$\lvert z\rvert > \lvert a\rvert$
7	$\cos(\Omega n+\theta)\,u[n]$	$\leftrightarrow$	$\frac{z^2\cos(\theta)-z\cos(\Omega-\theta)}{z^2-2z\cos(\Omega)+1}$	$\lvert z\rvert > 1$
7a	$2\lvert a\rvert^n\cos(\Omega n+\theta)\,u[n]$	$\leftrightarrow$	$\frac{ze^{j\theta}}{z-a}+\frac{ze^{-j\theta}}{z-a^*}$, $\quad a=\lvert a\rvert e^{j\Omega}$	$\lvert z\rvert > \lvert a\rvert$